1. Research question and motivation of the selected paper

1.1 Research Question

Did New Jersey's minimum wage increase on April 1, 1992 -- from $4.25 to $5.05 per hour, an 18.8% rise -- reduce employment at fast-food restaurants?

1.2 Motivation

The textbook competitive model yields an unambiguous prediction: a binding minimum wage raises labor costs and reduces employment (Stigler, 1946). Aggregate time-series studies of teenage employment from the 1970s generally confirmed this prediction.

However, several more-recent comparative studies, which contrast affected versus unaffected establishments or markets, fail to detect negative employment effects (Katz and Krueger, 1992; Card, 1992a, 1992b; Machin and Manning, 1994).

These conflicting empirical findings call for further credible establishment-level evidence derived from well-identified natural-experiment settings (Card and Krueger, 1994).

2. Data sources, sample construction, and variable definitions

2.1 Data sources

The primary dataset originates from the original authors' custom-designed telephone survey.

The authors conducted two survey waves targeting fast-food restaurants in New Jersey and eastern Pennsylvania:

- Wave I in February-March 1992 (shortly before New Jersey's minimum-wage hike).
- Wave II in November-December 1992 (approximately eight months after the policy change).

The sample includes restaurants from four chains: Burger King, KFC, Wendy's, and Roy Rogers.

2.2 Sample Construction

The initial sample consisted of 410 restaurants that completed the first survey wave, including 331 restaurants in New Jersey and 79 in Pennsylvania.

The authors subsequently attempted to interview all 410 restaurants again and obtained 399 second-wave interviews. They also collected information on restaurant closures.

The main employment regressions use a smaller estimation sample than the initial survey sample because some observations have missing data. For example, the paper's reduced-form employment regressions use 357 restaurants with employment and starting-wage information in both survey waves. The exact sample size therefore depends on the specification.

2.3 Variable Definitions

The main outcome variable is full-time-equivalent (FTE) employment:

$$FTE = EMPFT + NMGRS + 0.5 \times EMPPT$$

The principal explanatory variables are a New Jersey state dummy and the initial wage-gap measure (GAP). GAP captures the proportional increase in the starting wage required for a store to meet the new statutory minimum wage. Starting wage (dollars per hour):

wave-1	WAGE_ST
wave-2	WAGE_ST2

One secondary outcome is the tax-inclusive full-meal price, constructed as the sum of prices for a medium soda, small fries, and main entrée:

$$Full - mealprice = PSODA + PFRY + PENTREE$$

3. Empirical design

The paper employs a difference-in-differences strategy to estimate the effect of New Jersey's minimum-wage increase.

New Jersey restaurants constitute the treatment group, while restaurants in eastern Pennsylvania serve as the comparison group.

The authors adopt two complementary comparisons: the cross-state comparison between New Jersey and Pennsylvania, and an internal comparison within New Jersey, where high-wage stores act as a benchmark for low-wage stores.

3.1 Difference-In-Differences (DID)

The basic difference-in-differences estimator takes the form:

$$\Delta E = (\Delta E_{NJ}) - (\Delta E_{PA})$$

where ΔE_{NJ} is the average change in FTE employment for New Jersey restaurants, and ΔE_{PA} is the average change in FTE employment for eastern Pennsylvania restaurants.

This approach removes common changes affecting both groups, provided that the parallel-trends assumption holds in the absence of the policy.

3.2 Regression-Based Specifications

The authors also estimate regression models in which the dependent variable ΔE_i is the change in FTE employment of store i between the two survey waves.

One specification uses a New Jersey indicator NJ_i to measure the relative employment change:

$$\Delta E_i = \alpha + \beta\, NJ_i + \gamma X_i + \varepsilon_i$$

An alternative specification replaces the New Jersey dummy with the initial wage-gap variable GAP_i, a continuous measure of wage-shock intensity. GAP_i is defined as the proportional wage increase required for store i to meet the new statutory minimum wage of 5.05:

$$GAP_i = \begin{cases} 0, if store\ is\ in\ Pennsylvania, or\ W_{1i} \geq 5.05 \\ \dfrac{5.05 - W_{1i}}{W_{1i}}, otherwise \end{cases}$$

where W_{1i} is the starting wage of store i in Wave I.

The corresponding regression is:

$$\Delta E_i = \alpha' + \theta\, GAP_i + \gamma' X_i + \varepsilon'_i$$

The regressions include controls for restaurant chain and ownership status contained in vector $\boldsymbol{X}_i$, with additional specifications incorporating regional indicators.

The identification strategy relies on the assumption that, without the policy change, employment at restaurants in New Jersey would have evolved similarly to employment at comparable restaurants in Pennsylvania.

The authors conduct several specification checks, although the two-wave establishment survey does not by itself permit a direct test of pre-treatment parallel trends.

4. Core replication results and comparison with the original paper

4.1 Replication of Table 2

state	fte1	pct_ft1	wage_st	meal1	hrsopen	fte2	pct_ft2	wage_st2	meal2	hrsopen2
0	**23.331**	35.038	**4.630**	3.042	14.525	**21.166**	**30.385**	**4.617**	**3.027**	**14.654**
	11.856	23.933	0.352	0.601	2.951	8.277	24.463	0.357	0.566	2.888
	77.000	77.000	76.000	76.000	79.000	77.000	76.000	71.000	71.000	78.000
1	**20.439**	32.849	**4.612**	3.351	14.418	**21.027**	**35.866**	**5.081**	**3.415**	**14.420**
	9.106	23.750	0.346	0.644	2.779	9.293	24.454	0.105	0.636	2.721
	321.000	321.000	314.000	311.000	331.000	319.000	314.000	318.000	305.000	321.000
Total	20.999	33.272	4.616	3.290	14.439	21.054	34.798	4.996	3.341	14.466
	9.750	23.771	0.347	0.647	2.810	9.094	24.520	0.253	0.641	2.752
	398.000	398.000	390.000	387.000	410.000	396.000	390.000	389.000	376.000	399.000

Table 2 replicates closely.

In Wave I, full-time-equivalent (FTE) employment averages 20.4 workers in New Jersey and 23.3 in Pennsylvania, matching the paper's 20.4 and 23.3.

By Wave II, FTE employment is 21.0 in New Jersey and 21.2 in Pennsylvania (paper: 21.0 and 21.2).

The average starting wage in New Jersey rises from $4.6 in Wave I to $5.1 in Wave II, while the Pennsylvania starting wage remains essentially unchanged at $4.6.

After the increase, 85.5% of New Jersey stores pay exactly the new $5.05 minimum, compared with 1.3% in Pennsylvania (paper: 85.2 and 1.3).

The remaining Wave-II means -- the share of full-time workers (35.9 versus 30.4), the full-meal price (3.4 versus 3.0), weekday hours open (14.4 versus 14.7), and recruiting-bonus incidence (20.3 versus 23.4) -- all match the paper to one decimal place.

4.2 Replication of Table 3

Table 3 contains the paper's central result.

Mean FTE employment falls by 2.17 workers in Pennsylvania between the two survey waves but rises by 0.59 workers in New Jersey.

The difference-in-differences estimate is therefore +2.75 FTE workers (heteroskedasticity-robust standard error 1.34, t = 2.05), compared with +2.76 (t = 2.03) in the paper.

Restricting the sample to stores with valid employment data in both waves leaves the estimate essentially unchanged, and treating the four temporarily closed stores as having zero Wave-II employment yields a slightly smaller estimate, exactly as the paper reports.

Within New Jersey, employment expands at low-wage stores (those paying exactly $4.25 in Wave

I) and contracts at high-wage stores (those paying at least $5.00); the low-minus-high difference in the employment change is +3.36 FTE workers (t = 2.58).

4.3 Replication of Table 4

Table 4 estimates the same relationship in regression form, using the 357 stores with valid employment and starting-wage data in both waves.

Column (i) regresses the change in FTE employment on a New Jersey dummy, yielding a coefficient of 2.326 (standard error 1.19), versus 2.33 (1.19) in the paper.

Adding chain and company-ownership controls leaves the estimate at 2.304 (1.20) (paper: 2.30, 1.20). Columns (iii)-(v) replace the New Jersey dummy with the wage-gap variable GAP, which measures the proportional wage increase required to meet the new minimum.

The GAP coefficient is 15.653 (6.08) without controls, 14.916 (6.21) with chain and ownership controls, and 11.979 (7.42) with region dummies, against 15.65, 14.92, and 11.91 in the paper.

The dependent variable has a mean of -0.238 and a standard deviation of 8.825 (paper: -0.237 and 8.825), and the mean GAP among New Jersey stores is 0.105 (paper: 0.11).

4.4 Replication of Figure 1

Figure 1 reproduces the paper's four histograms of starting wages, plotted separately for the two states and two survey waves.

In Wave I, the New Jersey and Pennsylvania distributions are nearly identical.

By Wave II, the New Jersey distribution collapses onto the new $5.05 minimum -- 85.5 %t of stores pay exactly $5.05 -- while the Pennsylvania distribution remains essentially unchanged.

This first-stage pattern confirms that the April 1992 increase "bit" almost exclusively in New Jersey, which validates the use of Pennsylvania as the comparison group.

4.5 Comparison and Explanation of Differences

The replication matches the paper's published values to within rounding in every table.

The only non-negligible discrepancy is column (v) of Table 4 (11.98 versus 11.91), a 0.6% difference. Because the GAP coefficient in that specification is invariant to the choice of the omitted region category, this difference does not reflect an implementation choice but rather a trivial difference in the underlying data archive for a handful of stores.

None of the differences affects the paper's qualitative conclusion: the minimum-wage increase did not reduce fast-food employment.

5. Independent Extension

5.1 Economic Question and Motivation

This extension examines whether the employment effect of the minimum-wage increase varies with store size.

The question is motivated by competing theoretical predictions. The competitive model predicts a uniform reduction in employment that is independent of establishment size. Monopsony and search models, by contrast, predict that larger employers -- which face an upward-sloping labour-supply schedule and therefore hold greater labour-market power -- experience the largest employment gains following a binding minimum wage.

Store size thus provides a testable source of heterogeneity that distinguishes the two interpretations discussed in the original paper.

5.2 Empirical Approach

Store size is measured by Wave-I full-time-equivalent employment.

Two complementary specifications are estimated:

- The first splits the sample at the median of Wave-I FTE employment (19.5 workers) and estimates the difference-in-differences effect separately for large and small stores.
- The second interacts the New Jersey dummy with the continuous Wave-I FTE employment, exploiting the full variation in store size.

Every specification includes a main effect for store size to absorb mean reversion, and all standard errors are heteroskedasticity-robust.

5.3 Results

The results reveal a pronounced size gradient.

In the median split, large stores exhibit a difference-in-differences estimate of +4.94 FTE workers, whereas small stores exhibit an estimate of -0.29 FTE workers; the interaction is 5.23 (standard error 2.39, t = 2.19).

The continuous specification confirms this pattern: the interaction between the New Jersey dummy and Wave-I FTE employment is 0.26 (standard error 0.08, t = 3.30), implying that the employment effect rises by roughly 2.6 FTE workers for each one-standard-deviation increase in initial store size.

The positive employment effect is therefore concentrated entirely among larger stores.

5.4 Interpretation

This heterogeneity is difficult to reconcile with the competitive model, which predicts a size-invariant reduction in employment.

It is instead consistent with the monopsony and search interpretations discussed in the paper: larger establishments, facing an upward-sloping labour-supply schedule, expand employment in response to the minimum wage, while smaller establishments operating in more competitive

labour markets show no detectable employment gain.

The finding therefore provides independent evidence in favour of the monopsony interpretation and reinforces the paper’s conclusion that the minimum-wage increase did not reduce fast-food employment.

6. Validation, limitations, and conclusion

6.1 Validation

Several checks support the findings.

First, the replication matches the paper's published values to within rounding for every table, which confirms that the data construction and empirical implementation follow the original design.

Second, the difference-in-differences estimate is robust across sample definitions: the estimates using all available observations and the balanced subsample are virtually identical, and treating the temporarily closed stores as having zero Wave-II employment changes the result only slightly.

Third, the within-New-Jersey comparison between low-wage and high-wage stores yields an employment effect of similar magnitude to the cross-state comparison, which validates the use of Pennsylvania as the control group.

Fourth, the analysis is independently replicated in both Python and Stata, yielding consistent estimation results across platforms as a safeguard against coding errors. Both the Stata and Python implementations are included in the reproducibility package.

Finally, the store-size heterogeneity is robust across two specifications, the median split and the continuous interaction.

6.2 Limitations

Several limitations qualify these results.

First, the data contain only two survey waves, so the parallel-trends assumption underlying the difference-in-differences design cannot be directly tested; identification instead relies on the pre-treatment similarity of the two states, which is supported by the nearly identical Wave-I starting-wage distributions.

Second, the Pennsylvania comparison group is small (79 stores), and the within-chain subsamples are smaller still, which limits the precision of the heterogeneity estimates.

Third, the establishment-level data cannot capture the effect of the minimum wage on store entry and exit.

Fourth, the course-provided data archive differs trivially from the authors' original data, which accounts for the very small discrepancies such as the Table 4 column (v) coefficient.

Finally, employment is measured roughly eight months after the increase, so longer-run effects are not observed.

6.3 Conclusion

This project replicates the central finding of Card and Krueger (1994) and extends it.

The replication reproduces the paper's Tables 2–4 and Figure 1 almost exactly, confirming that New Jersey's minimum-wage increase did not reduce fast-food employment; if anything, employment rose by 2.75 FTE workers relative to Pennsylvania.

The independent extension shows that this positive employment effect is concentrated among larger stores, a pattern consistent with monopsony and search models and inconsistent with the competitive model.

Taken together, the results reinforce the paper's conclusion that the conventional prediction of minimum-wage-induced employment losses does not hold in this setting.

7. Appendix

Comparison of key results: original paper, Python, and Stata

Result	Paper	Python	Stata
FTE, Wave I (NJ / PA)	20.4 / 23.3	20.44 / 23.33	20.44 / 23.33
FTE, Wave II (NJ / PA)	21.0 / 21.2	21.03 / 21.17	21.03 / 21.17
Starting wage, Wave II (NJ / PA)	5.08 / 4.62	5.08 / 4.62	5.08 / 4.62
DiD (FTE workers)	2.76 (t = 2.03)	2.75 (t = 2.05)	2.75 (t = 2.05)
Table 4 (i): NJ dummy	2.33 (1.19)	2.326 (1.19)	2.326 (1.19)
Table 4 (ii): NJ + chain/ownership	2.30 (1.20)	2.304 (1.20)	2.304 (1.20)
Table 4 (iii): GAP	15.65 (6.08)	15.653 (6.08)	15.653 (6.08)
Table 4 (iv): GAP + chain/ownership	14.92 (6.21)	14.916 (6.21)	14.916 (6.21)
Table 4 (v): GAP + controls + region	11.91 (7.39)	11.979 (7.42)	11.979 (7.42)
Extension: large vs small DiD	—	+4.94 vs −0.29	+4.94 vs −0.29
Extension: continuous interaction	—	0.26 (t = 3.30)	0.26 (t = 3.30)